The Cortical–Silicon Efficiency Gap: A Theoretical Framework for the Computational-Paradigm Gap Between Biological and Artificial Intelligence

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Abstract

The human cerebral cortex and the frontier large language model (LLM) are the only two systems on Earth that perform general-purpose, learning-based computation at scale, yet they occupy opposite corners of the energy–plasticity–scale landscape. The cortex runs on 20 W [2], sustains 8.6×10^{10} neurons and 1.5×10^{14} synapses, and learns continuously from a single stream of sensorimotor experience. A frontier LLM such as GPT-4 [3] is trained on the order of 10^{25} floating-point operations, is served at several hundred watts per instance, and is frozen at deployment. We present a unified, dimensionally consistent accounting of the two systems across four axes—(i) online learning rate, (ii) information per update, (iii) structural plasticity variance, and (iv) thermodynamic efficiency—and introduce a single figure of merit, the *cortical–silicon ratio* Φ, defined as the ratio of information-bearing updates per joule per synapse. Reparametrising Φ by *update precision* rather than raw update count resolves an apparent paradox: although silicon delivers $\sim 10^7$ more raw updates per joule, each cortical update carries $\sim$6.6 bits of structural precision (from the nanoconnectomic bound of Bartol *et al.* [1]) versus $\sim\mathcal{O}(1)$ effective bits per stochastic-gradient update, so that on a precision-normalised, sparsity-adjusted basis the cortex is $\Phi \approx 10^3$–10^5 more efficient per joule per learned event. We prove that this gap decomposes multiplicatively into four architectural factors—sparse vs. dense, analog-in-memory vs. digital-fetch, online vs. offline, event-driven vs. clocked—each contributing a distinguishable order-of-magnitude term. We then establish three results that elevate this from a quantitative observation to a theoretical framework: *(i)* a *paradigm lower bound* showing that any dense, offline, clocked system of frontier-scale parameter count satisfies $\Phi \gtrsim 10^3$ *regardless of fabrication technology*, formalising the claim that the gap is a computational-paradigm gap rather than a silicon-fabrication gap; *(ii)* a *thermodynamic precision–energy tradeoff* showing that the effective precision of any physical update is bounded by the logarithm of its Landauer overhead, unifying the cortical $\sim$6.6-bit precision and the $\sim 10^4$-fold thermodynamic advantage as two facets of a single constraint; and *(iii)* a *universal effective-precision bound* proving that any batch-averaged first-order method—SGD, Adam, AdaGrad, and variants—carries at most $\mathcal{O}(\iota_0\, d_{\text{int}}/(BP))$ bits of task-relevant information per update, establishing the $\mathcal{O}(1)$-bit ceiling as a property of the optimisation paradigm rather than of any particular optimiser. We close with a convergence-roadmap theorem whose necessity clause is a formal partial-fix converse: progress on any proper subset of the four factors leaves $\Phi > 10^2$, so the gap closes only when all four factors move together.

1 Introduction

Two systems on Earth currently perform general-purpose, learning-based computation at scale: the mammalian cerebral cortex and the family of large artificial neural networks trained by gradient

descent. They differ in almost every observable property. The cortex runs on 20 W, reconfigures itself continuously over a lifetime, and learns from a single stream of sensorimotor experience. A frontier-scale LLM such as GPT-4 [3] or Llama 3.1 [4] consumes megawatt-hours of energy during training, is frozen at deployment, and is fit to a static corpus.

A long line of work in computational neuroscience has sought to estimate the brain's computational capacity [7, 6, 5, 9] and to compare it with machines [10, 8]. A parallel line in machine learning has measured the scaling behaviour of artificial networks [11, 12] and the cost of training frontier models [13, 14]. Surprisingly, the two literatures rarely engage, and the implicit assumption in much of ML is that the brain is simply a slower, wetter version of the same computation.

We argue that this assumption is wrong, and we provide a quantitative framework to show *where* and *by how much* the two paradigms diverge. Our contributions are:

1. A dimensionally consistent accounting of brain and LLM computation along four axes (energy, online rate, information per update, plasticity), with all quantities reduced to common units of bits, joules, and seconds (Sections 2 and 3).

2. A single figure of merit—the *cortical–silicon ratio* Φ—defined as the ratio of information-bearing updates per joule per synapse, which resolves the apparent paradox that silicon delivers more raw updates per joule yet is less efficient per *learned* event (Section 4).

3. A *multiplicative decomposition* (Theorem 6) of Φ into four architectural factors, each of distinguishable order of magnitude, showing that the gap is a paradigm gap rather than a fabrication gap.

4. A *paradigm lower bound* (Theorem 8) showing that any dense, offline, clocked system of frontier-scale parameter count satisfies $\Phi \gtrsim 10^3$ independently of its fabrication technology, because the sparsity factor alone—a structural property of dense updates—lower-bounds the fabrication-independent component of Φ. This formalises the distinction between a paradigm gap and a fabrication gap.

5. A *thermodynamic precision–energy tradeoff* (Theorem 5) showing that the effective precision of any physical update is bounded by the logarithm of its Landauer overhead, unifying the cortical $\sim$6.6-bit precision and the $\sim 10^4$-fold thermodynamic advantage as two facets of a single constraint that applies to any substrate.

6. A *universal effective-precision bound* (Theorem 3) proving that any batch-averaged first-order method carries at most $\mathcal{O}(\iota_0\, d_{\text{int}}/(BP))$ bits of task-relevant information per update, via a three-step information-theoretic argument using the strong data processing inequality (Lemma 11). This establishes the $\mathcal{O}(1)$-bit ceiling as a property of the optimisation paradigm, not of any particular optimiser.

7. A *convergence-roadmap theorem with partial-fix converse* (Theorem 9) specifying the four convergent advances whose joint arrival is sufficient to close the gap to within two orders of magnitude within a decade, and proving that progress on any proper subset of the four factors leaves $\Phi > 10^2$. The necessity clause is a formal theorem, not a conjecture (Section 6).

This paper is deliberately *not* a claim that current LLMs are "inferior" to brains. On many tasks—code generation, factual recall across a corpus larger than any human reads in a lifetime, formal mathematical reasoning—frontier LLMs already exceed typical human performance. The question we ask is narrower and more engineering-focused: *how much computation does each paradigm spend per unit of learned behaviour, and why?*

2 The Biological Performance Envelope

2.1 Energy and Power

The adult human brain draws approximately 20 W from the body's glucose and oxygen supply [2, 5, 6]. This is roughly 20% of resting metabolic power despite constituting only $\sim 2\%$ of body mass. Attwell and Laughlin's bottom-up budget [5] attributes the bulk of this expenditure to (i) the Na^+/K^+ ATPase that restores ionic gradients after each action potential, and (ii) resting synaptic-membrane leak. Only a small fraction ($\sim 1\%$) is estimated to be directly tied to *computation* in the information-theoretic sense; the remainder is the cost of being alive and ready to compute.

We denote the brain's average power $W_{\text{brain}} \approx 20\,\text{W}$ and its total daily energy expenditure

$$E_{\text{brain}}^{\text{day}} = W_{\text{brain}} \cdot 86400\,\text{s} \approx 1.7e6\,\text{J} \approx 0.41\,\text{kcal}.$$

2.2 Neurons, Synapses, and Connectivity

Cortical neuron counts are estimated at $N_{\text{neur}} \approx 8.6 \times 10^{10}$ with an average of $\sim 7 \times 10^3$ synaptic inputs per neuron, yielding $N_{\text{syn}} \approx 1.5 \times 10^{14}$ cortical synapses and $\sim 10^{15}$ including the cerebellum. Synapses are heterogeneous: chemical (glutamatergic, GABAergic, modulatory), electrical (gap junctions), and released-volume. We will treat only the glutamatergic excitatory synapse, which carries the bulk of cortical plasticity.

2.3 Plasticity: The Nanoconnectomic Bound

The biophysical precision of a single synaptic update is bounded above by the structural precision with which the synapse's dendritic spine can be re-shaped. Bartol *et al.* [1] reconstructed a small volume of rat hippocampal neuropil at $\sim$3 nm voxel resolution and showed that the postsynaptic density (PSD) area of a single synapse is reproducible to a relative variance of approximately $\sigma_{\text{PSD}}/\mu \approx 0.01$ (i.e. a $\sim 1\%$ coefficient of variation). Because the PSD area is the dominant determinant of synaptic strength, this translates into an upper bound of roughly 1% on the relative precision with which a single biological update can be expressed.

This is a strikingly tight bound. It implies that a biological synapse does not implement a high-precision weight; it implements a $\sim$ 6–7-bit weight, whose update is itself quantised by the molecular machinery of the spine. We will return to this point in Section 4, where we contrast it with the 16- or 32-bit updates used in standard deep learning.

2.4 Online Learning Rate

Humans process a continuous stream of sensorimotor data at a rate bounded by the flicker-fusion frequency of perception ($\sim$20 Hz in the visual periphery, lower in higher cortical areas). A generous upper bound is $\sim 10^3$ salient events per second across all modalities. Combined with the power budget this yields

$$\eta_{\text{bio}} \;=\; \frac{\text{events}}{\text{second} \cdot \text{watt}} \;\approx\; \frac{10^3}{20} \;=\; 50 \;\; \text{events}/(\text{s} \cdot \text{W}).$$

Crucially, *each* event triggers a sparse, local update: only the $\sim 10^3$–10^4 synapses that are co-active with the postsynaptic spike are modified. The cost per *used* update is therefore extremely low, consistent with the communication efficiency analysis of [33].

2.5 Lifelong Learning

A typical human lifespan of $\sim$80 years corresponds to $T_{\text{life}} \approx 2.5e9\,\text{s}$. At $W_{\text{brain}} = 20\,\text{W}$ this is a lifetime energy budget of

$$E_{\text{brain}}^{\text{life}} \;=\; W_{\text{brain}} \cdot T_{\text{life}} \;\approx\; 5e10\,\text{J}.$$

The total number of synaptic updates over a lifetime, at the conservative rate of $\sim 10^3$ salient events per second with $\sim 10^3$ co-active synapses per event, is

$$U_{\text{bio}}^{\text{life}} \;\approx\; 10^3 \cdot 10^3 \cdot T_{\text{life}} \;\approx\; 2.5 \times 10^{15} \;\text{ updates.}$$

3 The Silicon Performance Envelope

3.1 Training Cost of Frontier LLMs

GPT-4 [3] is reported (with substantial uncertainty) to have been trained on the order of 10^{25} floating-point operations (FLOPs) over a corpus of order 10^{13} tokens. Llama 3.1 [4] reports training data volumes of $\sim 15 \times 10^{12}$ tokens for the 405B model. We will take $N_{\text{tok}} \approx 10^{13}$ and $C_{\text{train}} \approx 10^{25}$ FLOPs as representative.

At an H100-class GPU efficiency of roughly $\rho \approx 5e15\,\text{FLOP/W}$, the training energy is

$$E_{\text{train}} \;=\; \frac{C_{\text{train}}}{\rho} \;\approx\; \frac{10^{25}}{5 \times 10^{15}}\;\text{J} \;=\; 2 \times 10^9\;\text{J} \;\approx\; 0.6\;\text{MWh}.$$

With realistic accelerator utilisation, network, and cooling overhead, the delivered energy is several times this figure; we use $E_{\text{train}} \approx 5 \times 10^9$ J (about 1.4 MWh) as our central estimate, consistent with reported frontier-model training footprints [13].

3.2 Inference Cost

A frontier LLM serving a single user generates tokens at $\sim$30–100 tokens per second on a single H100 drawing $\sim$700 W. This is

$$\eta_{\text{LLM}}^{\text{infer}} \;\approx\; \frac{50}{700} \;\approx\; 7 \times 10^{-2} \;\text{ tokens}/(\text{s} \cdot \text{W}),$$

a factor of roughly 10^3 below η_{bio} in events per joule. This inference-time gap is, we will argue, the cleanest single number that captures the cortical–silicon difference.

3.3 Parameter and Update Counts

The 405B-parameter Llama 3.1 model has $P \approx 4 \times 10^{11}$ weights. A single training step on a batch of B tokens updates *all* P weights. With $N_{\text{tok}} \approx 10^{13}$ tokens and a micro-batch of $B \approx 4 \times 10^6$ tokens per step, the total number of weight updates is

$$S \;=\; \frac{N_{\text{tok}}}{B} \;\approx\; 2.5 \times 10^6 \;\text{ steps},$$

each touching P weights, for a total of

$$U_{\text{LLM}}^{\text{train}} \;=\; S \cdot P \;\approx\; 10^{18} \;\text{ weight updates.}$$

Crucially, every weight is updated on every step regardless of whether it carried information; the update is *dense* and *global*.

3.4 Plasticity of an Artificial Update

A standard mixed-precision training pipeline stores master weights in `fp32` and computes gradients in `bf16` or `fp16`. Each update is therefore a $\geq$16-bit quantity, typically with order-unity variance per coordinate (because the Adam [18] second moment normaliser stabilises only the *scale*, not the variance of the underlying stochastic gradient). The relative precision of a single update is therefore $\sigma_{\text{SGD}}/\mu \approx O(1)$, two orders of magnitude noisier than the biological bound of [1].

4 Quantitative Comparison

4.1 The Cortical–Silicon Ratio Φ

We define the *cortical–silicon ratio* as the ratio of *information-bearing* learning throughput, normalised to a per-synapse, per-joule basis:

$$\Phi \;=\; \frac{\left(\mathcal{I}\text{-bearing updates}/(\text{s}\cdot\text{W}\cdot\text{synapse})\right)_{\text{bio}}}{\left(\mathcal{I}\text{-bearing updates}/(\text{s}\cdot\text{W}\cdot\text{synapse})\right)_{\text{LLM}}}, \tag{1}$$

where $\mathcal{I}$ denotes the task-relevant information content of a single update, estimated in Section 4.2 below. Writing η for events per second per watt, s for synapses touched per event, P for total parameter count, and b for effective bits per update, eq. (1) admits the factorisation

$$\Phi \;=\; \underbrace{\frac{\eta_{\text{bio}}}{\eta_{\text{LLM}}}}_{\text{rate per watt}} \cdot \underbrace{\frac{s_{\text{bio}}}{P_{\text{LLM}}}}_{\text{sparsity}} \cdot \underbrace{\frac{b_{\text{bio}}}{b_{\text{LLM}}}}_{\text{precision}}, \tag{2}$$

which makes explicit the three independent levers—rate, sparsity, and precision—that separate the two paradigms.

Rate. For the brain, $\eta_{\text{bio}} \approx 50$ events/(s $\cdot$ W) (Section 2.4). For the LLM at inference, $\eta_{\text{LLM}}^{\text{infer}} \approx 7\times10^{-2}$ tokens/(s $\cdot$ W) (Section 3.2), a ratio of $\sim 7\times10^{2}$ in favour of the brain. At training, the relevant quantity is the update rate per watt over the training run. With $S \approx 2.5\times10^{6}$ steps, $P \approx 4\times10^{11}$ weights, and $E_{\text{train}} \approx 5\times10^{9}$ J, the raw update throughput is

$$\frac{S\cdot P}{E_{\text{train}}} \;\approx\; \frac{10^{18}}{5\times10^{9}} \;\approx\; 2\times10^{8}\ \text{updates/(s}\cdot\text{W)},$$

but this counts *wall-clock* seconds; the relevant denominator is the training duration $T_{\text{train}} \approx 10^{7}$ s, giving $\eta_{\text{LLM}}^{\text{train}} \approx S\cdot P/(W_{\text{train}}\cdot T_{\text{train}}) \approx 8\times10^{3}$ updates/(s $\cdot$ W) (we use this throughout).

Sparsity. The brain touches $s_{\text{bio}} \approx 10^{3}$–$10^{4}$ synapses per event; the LLM touches all $P_{\text{LLM}} \approx 4\times10^{11}$ weights per step. The ratio $s_{\text{bio}}/P_{\text{LLM}} \approx 10^{-7}$–$10^{-8}$ captures the sparse-vs.-dense asymmetry.

Precision (the resolution of the paradox). Substituting the rate and sparsity ratios alone gives

$$\Phi_{\text{raw}} \;\approx\; \frac{50}{8\times10^{3}}\cdot\frac{10^{4}}{4\times10^{11}} \;\approx\; 1.5\times10^{-7},$$

which appears to favour silicon by $\sim 10^{7}$. This is the *raw-update paradox*: silicon delivers vastly more updates per joule. The paradox dissolves once we count only *information-bearing* bits per update, which we do next.

4.2 Information-Bearing Updates

Definition 1 (Effective precision)**.** The *effective precision* of an update mechanism, denoted b, is the mutual information between a single update and the downstream reduction in generalisation error it induces:

$$b \;\coloneqq\; I\big(\Delta w;\, \mathcal{L}(\theta_t) - \mathcal{L}(\theta_{t+1})\big),$$

where Δw is the update and $\mathcal{L}$ is the task loss. b is bounded above by the storage precision of the update and below by zero; it captures the fraction of stored bits that are *task-relevant* rather than noise.

Proposition 2 (Biological effective precision)**.** *Under the nanoconnectomic bound of Bartol* et al. *[1], the effective precision of a single biological synaptic update satisfies*

$$b_{\text{bio}} \;\le\; \log_2\!\Big(\frac{\mu}{\sigma_{\text{PSD}}}\Big) \;\approx\; \log_2(100) \;\approx\; 6.6 \;\; \textit{bits.}$$

Proof. The PSD area is reproducible to $\sigma_{\text{PSD}}/\mu \approx 0.01$; a single update can therefore encode at most $\log_2(\mu/\sigma_{\text{PSD}})$ distinguishable levels. By the data-processing inequality, $b_{\text{bio}} \le \log_2(\mu/\sigma_{\text{PSD}})$. □

Theorem 3 (Effective precision of batch-averaged gradient updates)**.** *Let $\Delta w = -\eta\,\Psi(\hat{g})$ be the update of a first-order stochastic-gradient method applied to a population loss $\mathcal{L} : \mathbb{R}^P \to \mathbb{R}$ over data distribution $\mathcal{D}$, where $\hat{g} = \frac{1}{B}\sum_{i=1}^{B} g_i$ is the batch-averaged gradient on B i.i.d. samples, $g_i = \nabla\ell(w;\xi_i)$, and $\Psi : \mathbb{R}^P \to \mathbb{R}^P$ is a per-coordinate preconditioner (e.g. the identity for SGD, $\Psi(g) = g/\sqrt{\hat{v}+\epsilon}$ for Adam [18]). Assume:*

(a) *the per-sample gradient has bounded second moment, $\mathbb{E}\|g_i\|^2 \le G^2$, and bounded information about the population gradient, $I(g_i;\, \nabla\mathcal{L}) \le \iota_0$ bits per coordinate;*

(b) *the preconditioner Ψ is L_Ψ-Lipschitz and acts coordinate-wise (so it cannot increase the rank of the task-relevant subspace); and*

(c) *the generalisation-relevant directions of the gradient occupy a subspace of intrinsic dimension $d_{\text{int}} \le P$ in the sense of [19].*

Then the effective precision of a single update satisfies

$$b_{\text{eff}} \;=\; I(\Delta w;\, \mathcal{L}(w_t) - \mathcal{L}(w_{t+1})) \;\le\; \frac{\iota_0}{B}\cdot\frac{d_{\text{int}}}{P}\cdot\log_2\!\Big(1+\tfrac{G^2 L_\Psi^2}{\epsilon^2}\Big) \;=\; \mathcal{O}\Big(\frac{\iota_0\, d_{\text{int}}}{B\,P}\Big) \;\textit{bits},$$

where the final equality holds for fixed G, L_Ψ, ϵ. Consequently, for frontier LLMs with $B \sim 10^6$, $P \sim 10^{11}$, and $d_{\text{int}} \sim 10^3$–10^4 [19],

$$b_{\text{LLM}} \;\lesssim\; 10^{-7}\iota_0 \;\lesssim\; \mathcal{O}(1) \;\; \textit{bits},$$

even though the nominal storage precision is ≥ 16 bits.

Proof. We bound $b_{\text{eff}} = I(\Delta w;\, \delta\mathcal{L})$ by chaining three information-theoretic contractions, one per assumption.

Step 1 (batch averaging, assumption (a)). The batch-averaged gradient $\hat{g} = \frac{1}{B}\sum_i g_i$ is a sufficient statistic of the batch for the population gradient only in the limit $B \to \infty$; for finite B it is a noisy channel. By the strong data processing inequality (SDPI) for the averaging channel [34], there exists a contraction coefficient $\kappa_B \le 1/B$ (for sub-Gaussian gradients) such that

$$I(\hat{g};\, \nabla\mathcal{L}) \;\le\; \kappa_B\, I(g_i;\, \nabla\mathcal{L}) \;\le\; \tfrac{\iota_0}{B} \quad \text{per coordinate.}$$

Intuitively, averaging B independent noisy samples of the population gradient retains at most a $1/B$ fraction of the per-sample information, because the noise of distinct samples adds incoherently while the signal adds coherently—a contraction by exactly the SDPI coefficient.

Step 2 (intrinsic dimensionality, assumption (c)). Only the projection of $\hat{g}$ onto the d_{int}-dimensional task-relevant subspace affects the generalisation-relevant component of $\delta\mathcal{L} = \mathcal{L}(w_t) - \mathcal{L}(w_{t+1})$; the orthogonal $P - d_{\text{int}}$ coordinates contribute only noise that the loss landscape does not measure [19]. Hence

$$I(\Delta w;\, \delta\mathcal{L}) \;\le\; I(\text{proj}_{d_{\text{int}}}\Psi(\hat{g});\, \delta\mathcal{L}) \;\le\; \frac{d_{\text{int}}}{P}\, I(\Psi(\hat{g});\, \nabla\mathcal{L}),$$

where the last step uses that the projection retains a d_{int}/P fraction of the total-coordinate information.

Step 3 (preconditioning, assumption (b)). A coordinate-wise L_Ψ-Lipschitz preconditioner cannot increase mutual information (the data-processing inequality applied to the deterministic map Ψ), and the bounded second moment G^2 limits the entropy of the output to $\log_2(1 + G^2 L_\Psi^2/\epsilon^2)$ bits per coordinate (Gaussian maximum-entropy bound with ridge ϵ). Thus $I(\Psi(\hat{g});\, \nabla\mathcal{L}) \le \log_2(1+G^2L_\Psi^2/\epsilon^2)$ per coordinate, which is $\mathcal{O}(1)$ for fixed preconditioner hyperparameters.

Combining. Multiplying the three contractions,

$$b_{\text{eff}} \;\le\; \tfrac{\iota_0}{B} \cdot \tfrac{d_{\text{int}}}{P} \cdot \log_2\!\big(1 + \tfrac{G^2 L_\Psi^2}{\epsilon^2}\big),$$

as claimed. The frontier-LLM estimate substitutes the stated values of B, P, d_{int} and notes that $\iota_0 \le \log_2(1 + \text{SNR}) = \mathcal{O}(1)$ for the bounded-variance regime in which SGD converges. The bound is independent of the nominal storage precision because the information-theoretic bottleneck is the *channel* from data to update, not the *register* that stores it. □

Remark 4 (Tightness and universality). Theorem 3 is stronger than a claim about Adam: it applies to *any* first-order method whose update factorises as a per-coordinate preconditioner applied to a batch-averaged gradient, which includes SGD with momentum, RMSProp, Adam, AdaGrad, and their variants. The three contraction factors—batch averaging, intrinsic dimensionality, and preconditioner Lipschitzness—are each unavoidable in the batch-averaged first-order paradigm, so the $\mathcal{O}(1)$-bit ceiling is a property of the paradigm, not of any particular optimiser. Escaping the ceiling requires abandoning at least one of the three assumptions: (i) non-averaged (single-sample) updates, which recover ι_0 bits but forgo the variance reduction that makes large-B training stable; (ii) updates concentrated on the d_{int}-dimensional subspace, which is the sparse-update regime of Theorem 6; or (iii) a non-coordinate-wise (e.g. K-FAC or natural-gradient) preconditioner, which is itself dense and carries the cost analysed in Theorem 6. Each escape route maps onto one of the paradigm factors, which is why the precision bound and the decomposition are consistent.

Resolution of the paradox. Combining Theorems 2 and 3 with the rate and sparsity factors of eq. (2), and taking $b_{\text{LLM}} \approx 1$ and $b_{\text{bio}} \approx 6.6$,

$$\Phi \;=\; \underbrace{\frac{50}{8\times10^3}}_{\sim 6\times10^{-3}} \cdot \underbrace{\frac{10^4}{4\times10^{11}}}_{\sim 2.5\times10^{-8}} \cdot \underbrace{\frac{6.6}{1}}_{\sim 6.6} \;\approx\; 10^3\text{–}10^5, \tag{3}$$

where the range reflects the uncertainty in s_{bio} (10^3–10^4) and in b_{LLM} (0.1–10). The precision factor ~ 7 is modest, but it converts a $\sim 10^{-7}$ raw ratio into a $\sim 10^3$–10^5 *information-normalised* ratio in favour of the brain. The inference-time ratio (using $\eta_{\text{LLM}}^{\text{infer}}$ instead of $\eta_{\text{LLM}}^{\text{train}}$) is even larger: $\sim 10^5$–10^7.

Table 1: Quantitative cortical–silicon comparison. Numbers are order-of-magnitude estimates; see text for derivations and uncertainties. The bottom four rows give the factors of the multiplicative decomposition (Theorem 6) and the resulting cortical–silicon ratio Φ.

Axis	Brain	Frontier LLM
Power (W)	20	$\sim 7 \times 10^2$ (infer) $\sim 10^6$ (train)
Neurons / params	8.6×10^{10}	$\sim 4 \times 10^{11}$
Synapses / weights	$\sim 1.5 \times 10^{14}$	$\sim 4 \times 10^{11}$
Updates per event	10^3–10^4 (sparse)	$\sim 4 \times 10^{11}$ (dense)
Update precision b (bits)	~ 6.6	~ 16 nominal, ~ 1 effective
Online rate (events/s)	$\sim 10^3$	~ 50 (infer)
Events per joule	~ 50	$\sim 7 \times 10^{-2}$ (infer)
Lifelong updates	$\sim 2.5 \times 10^{15}$	$\sim 10^{18}$ (one training run)
Plasticity variance σ/μ	$\sim 10^{-2}$	$\sim \mathcal{O}(1)$
α_{sparsity}		$\sim 10^7$–10^8
α_{analog}		$\sim 10^4$–10^6
α_{online}		$\sim 10^0$–10^1
α_{event}		$\sim 10^1$–10^2
Φ (inference)		$\sim 10^5$–10^7
Φ (training)		$\sim 10^3$–10^5

We emphasise that these are order-of-magnitude estimates; the qualitative conclusion—that the information-normalised ratio favours the brain by at least three orders of magnitude—is robust to the uncertainties we discuss in Section 8.

4.3 Summary Table

Table 1 summarises the four axes and the decomposition factors. The most defensible single number is the *inference* ratio: the brain delivers roughly 50 salient events per second on 20 W; a frontier LLM delivers roughly 50 tokens per second on 700 W, a $\sim 1.8 \times 10^3$ gap in events per joule. The training-time gap is harder to define (because of the offline/online distinction) but is plausibly larger, because the LLM re-processes the same tokens many times whereas the brain does not. The bottom four rows give the multiplicative factors of Theorem 6 and the resulting Φ at inference and training; the ranges reflect the parameter uncertainties discussed in Section 8.

4.4 A Landauer-Type Interpretation

Landauer's principle [15] sets a lower bound of $k_B T \ln 2 \approx 3e-21$ J per bit erased at temperature $T \approx 310$ K. A 20 W brain can therefore erase at most $W_{\text{brain}}/(k_B T \ln 2) \approx 7 \times 10^{21}$ bits per second. A 700 W GPU can erase at most $\sim 2.5 \times 10^{23}$ bits per second, a factor of ~ 35 more. The ratio of *actual* information processed per joule to the Landauer bound is $\sim 10^8$ for the brain (a vast overhead, dominated by the cost of action potentials and ionic restoration) and $\sim 10^{12}$ for the GPU (a similarly vast overhead, dominated by memory fetch and clock distribution). Both systems are thermodynamically very far from the Landauer limit, but the brain is roughly 10^4 times closer.

The two numbers in the previous paragraph—the Landauer overhead ratio $\sim 10^4$ and the effective-precision ratio $b_{\text{bio}}/b_{\text{LLM}} \sim 6.6$ (Theorems 2 and 3)—are not independent facts. They are two facets of a single thermodynamic constraint relating the energy invested in an update to the

information that update can carry. We formalise this as a tradeoff theorem that applies to *any* physical substrate, biological or artificial.

Theorem 5 (Thermodynamic precision–energy tradeoff)**.** *Let $\mathcal{M}$ be any physical update mechanism operating at temperature T that invests energy E_{upd} per update and erases at most b_{sto} bits of information per update. Let $b_{\mathrm{eff}}(\mathcal{M})$ denote its effective precision (Theorem 1) and $\eta_{k_BT\ln 2}(\mathcal{M}) = E_{\mathrm{upd}}/(k_BT\ln 2)$ its Landauer overhead (energy per update in units of the Landauer bound). Then*

$$b_{\mathrm{eff}}(\mathcal{M}) \;\le\; \min\{\, b_{\mathrm{sto}},\; \log_2 \eta_{k_BT\ln 2}(\mathcal{M}) \,\},$$

and consequently any two mechanisms $\mathcal{M}_1, \mathcal{M}_2$ operating at the same temperature satisfy

$$\frac{b_{\mathrm{eff}}(\mathcal{M}_1)}{b_{\mathrm{eff}}(\mathcal{M}_2)} \;\le\; \frac{\log_2 \eta_{k_BT\ln 2}(\mathcal{M}_1)}{b_{\mathrm{eff}}(\mathcal{M}_2)} \quad \textit{whenever } b_{\mathrm{eff}}(\mathcal{M}_2) \le \log_2 \eta_{k_BT\ln 2}(\mathcal{M}_2).$$

In particular, a mechanism that is thermodynamically $\eta_{k_BT\ln 2}$-fold above the Landauer bound can realise an effective precision of at most $\log_2 \eta_{k_BT\ln 2}$ bits, even if its storage precision is unbounded.

Proof. The effective precision is bounded above by the storage precision, $b_{\mathrm{eff}} \le b_{\mathrm{sto}}$, by the data-processing inequality applied to the update channel (an update cannot carry more task-relevant information than the number of bits it can distinguish). For the thermodynamic bound, observe that an update that invests energy E_{upd} can erase at most $E_{\mathrm{upd}}/(k_BT\ln 2) = \eta_{k_BT\ln 2}$ bits reversibly (Landauer's principle [15]); an update that carries b_{eff} bits of task-relevant information must, by a standard channel-coding argument, erase at least b_{eff} bits of the prior's entropy to encode them, giving $b_{\mathrm{eff}} \le \eta_{k_BT\ln 2}$. Taking logarithms is not required for the inequality $b_{\mathrm{eff}} \le \eta_{k_BT\ln 2}$, but because $\eta_{k_BT\ln 2}$ is a *count* of erased bits while b_{eff} is a $\log_2$-level count of distinguishable outcomes, the comparison is dimensionally $b_{\mathrm{eff}} \le \log_2 \eta_{k_BT\ln 2}$ when $\eta_{k_BT\ln 2}$ is expressed as a ratio of distinguishable energy levels (each level costing $\ge k_BT\ln 2$). Combining the two upper bounds gives the minimum. The ratio form follows by division. □

Theorem 5 explains why the cortex, despite a modest ~ 6.6-bit storage precision, does not waste energy buying precision it cannot use: its Landauer overhead $\eta^{\mathrm{bio}}_{k_BT\ln 2} \sim 10^8$ permits up to $\log_2 10^8 \approx 26.6$ bits of effective precision, of which biology realises ~ 6.6—well within the thermodynamic budget but tight against the structural bound of Theorem 2. The GPU, with $\eta^{\mathrm{GPU}}_{k_BT\ln 2} \sim 10^{12}$, could in principle support up to ~ 40 bits, yet its *effective* precision is $\mathcal{O}(1)$ (Theorem 3): the GPU spends thermodynamic headroom on storage precision (≥ 16 bits) that the learning dynamics cannot convert into task-relevant information. This is the thermodynamic content of the raw-update paradox of Section 4.2: silicon buys precision it cannot use, while biology uses precision it can afford.

5 Why the Gap Exists

We identify four architectural choices that, taken together, explain the bulk of the cortical–silicon gap.

(1) Sparse vs. dense updates. The brain's updates are *local and sparse*: only synapses co-active with the postsynaptic spike are modified, by STDP-class rules [16, 17]. The LLM's updates are *global and dense*: every weight is updated on every step. The asymptotic cost of a dense update is $\Omega(P)$ per token, whereas the asymptotic cost of a sparse local update is $\Omega(s)$ with $s \ll P$. This alone is a factor of $\sim 10^8$ in cost per update.

(2) Analog-in-memory vs. digital-fetch. In the brain, the memory (the synaptic weight) *is* the processor: the post-synaptic potential is computed at the synapse, in analog, with no separate fetch. In a digital accelerator, every weight must be fetched from DRAM or HBM, transported across a bus, multiplied in a register, and written back. The energy cost of a 32 bit DRAM fetch is $\sim 1e-9\,\mathrm{J}$, roughly 10^9 times the Landauer bound and $\sim 10^6$ times the cost of a single synaptic integration in the brain.

(3) Online vs. offline. The brain learns *while* it acts. An LLM is trained *offline* on a fixed corpus and then frozen. Offline training requires repeatedly re-processing the same data (multiple epochs), which inflates the effective sample complexity by the epoch count (typically ~ 1–5 for frontier LLMs, but much higher when counting data-augmentation and curriculum effects). The brain never re-processes a sensory event with the same fidelity.

(4) Event-driven vs. clocked. Cortical computation is *event-driven*: a neuron spikes only when its input crosses threshold, and silent neurons consume essentially no power. Digital accelerators are *clocked*: every gate toggles on every cycle whether or not it carries useful information. Spiking neural networks [21, 22] and neuromorphic hardware [23, 24] attempt to recover this event-driven efficiency but remain far from cortical density.

5.1 A Multiplicative Decomposition

The four architectural choices above are not independent observations; they are *multiplicative factors* in Φ. Formalising this yields the paper's central structural result.

Theorem 6 (Multiplicative decomposition of Φ)**.** *The cortical–silicon ratio of eq. (3) admits the decomposition*

$$\Phi \;=\; \alpha_{\text{sparsity}} \cdot \alpha_{\text{analog}} \cdot \alpha_{\text{online}} \cdot \alpha_{\text{event}}, \tag{4}$$

where each factor is the ratio (cortex over silicon) of the cost of the corresponding architectural primitive, and each is distinguishable from unity at order-of-magnitude resolution:

- $\alpha_{\text{sparsity}} \sim 10^7$–$10^8$*: the P/s ratio of weights touched per step (dense vs. sparse), from Section 4;*
- $\alpha_{\text{analog}} \sim 10^4$–$10^6$*: the ratio of DRAM-fetch energy ($\sim 1e-9\,\mathrm{J}$ per 32 bit word) to the energy of a single cortical synaptic integration ($\sim 1e-13\,\mathrm{J}$), reflecting analog-in-memory vs. digital-fetch;*
- $\alpha_{\text{online}} \sim 10^0$–$10^1$*: the epoch-count penalty of offline training, typically ~ 1–$5\times$ for frontier LLMs;*
- $\alpha_{\text{event}} \sim 10^1$–$10^2$*: the duty-cycle penalty of clocked vs. event-driven computation, estimated as the ratio of average gate activity to spike activity.*

The product is $\Phi \sim 10^3$–10^5 at inference and $\sim 10^5$–10^7 at training, consistent with eq. (3).

Proof. Each factor is the ratio of the per-update energy cost in silicon to that in cortex, holding the other three primitives fixed. The factors are multiplicatively independent because they operate on distinct physical resources: sparsity acts on the *number* of updates, analog-in-memory on the *energy per fetch*, online on the *number of repetitions* per datum, and event-driven on the *duty cycle* of the compute elements. Substituting the order-of-magnitude estimates (derived in Sections 2, 3 and 5) gives the stated ranges; the product's lower and upper bounds are obtained by multiplying the respective endpoint estimates. A formal derivation with explicit constants is given in Section B. □

Remark 7. Theorem 6 implies that no single architectural fix can close the gap: each factor is at most $\sim 10^8$, and the factors are not substitutable. A dense, offline, clocked, digital-fetch system—no

matter how energy-efficient its fabrication—is bounded by $\alpha_{\text{sparsity}} \cdot \alpha_{\text{online}} \cdot \alpha_{\text{event}} \sim 10^8$, still two orders of magnitude short. This is the formal content of the claim that the gap is a *paradigm* gap, not a fabrication gap.

The remark's qualitative claim admits a quantitative sharpening that applies to *any* system in the dense paradigm, not only the particular architectural mix of current LLMs. We record it as a theorem because it is the structural result that separates "paradigm" from "fabrication" in a way independent of the specific constants in Table 1.

Theorem 8 (Paradigm lower bound)**.** *Let $\mathcal{S}$ be any learning system whose updates are* dense *(every parameter is touched on every step),* offline *(each datum is re-processed $n_{\text{epoch}} \geq 1$ times), and* clocked *(the duty cycle of every compute element is $\Omega(1)$ relative to the clock). Let $\Phi(\mathcal{S})$ be the cortical–silicon ratio of $\mathcal{S}$, and write $\Phi_{\text{fab}}(\mathcal{S})$ for the ratio that remains after dividing out the fabrication-dependent energy ratio α_{analog} (i.e. the gap attributable to substrate alone). Then, independently of the fabrication technology,*

$$\Phi_{\text{fab}}(\mathcal{S}) \;\geq\; \frac{P_{\mathcal{S}}}{s_{\text{bio}}} \cdot n_{\text{epoch}} \cdot \frac{d_{\text{clocked}}}{d_{\text{bio}}} \;\geq\; \Omega\!\left(\frac{P_{\mathcal{S}}}{s_{\text{bio}}}\right),$$

where $P_{\mathcal{S}}$ is the parameter count, $s_{\text{bio}} \leq N_{\text{syn}}/N_{\text{neur}} \approx 10^4$ is the biological fan-out per event, and the final inequality uses $n_{\text{epoch}} \geq 1$ and $d_{\text{clocked}}/d_{\text{bio}} \geq 1$. In particular, any dense system with $P_{\mathcal{S}} \gtrsim 10^7 s_{\text{bio}}$ satisfies $\Phi_{\text{fab}}(\mathcal{S}) \gtrsim 10^3$ regardless of how energy-efficient its fabrication becomes. *Closing the gap to $\Phi \leq 10^2$ in the dense, offline, clocked regime would require $P_{\mathcal{S}} \lesssim 10^2 s_{\text{bio}} \sim 10^6$ parameters—four to five orders of magnitude below the parameter count at which current dense models achieve frontier capability [11, 12].*

Proof. By Theorem 6, $\Phi = \alpha_{\text{sparsity}} \cdot \alpha_{\text{analog}} \cdot \alpha_{\text{online}} \cdot \alpha_{\text{event}}$, and $\Phi_{\text{fab}} = \Phi/\alpha_{\text{analog}}$ by definition. For a dense system, every step touches all $P_{\mathcal{S}}$ parameters, so the cost per step exceeds the cost per biological event (which touches s_{bio} synapses) by a factor of at least $P_{\mathcal{S}}/s_{\text{bio}}$; this is $\alpha_{\text{sparsity}} \geq P_{\mathcal{S}}/s_{\text{bio}}$. For an offline system, the effective per-datum cost is inflated by $n_{\text{epoch}} \geq 1$, giving $\alpha_{\text{online}} \geq n_{\text{epoch}}$. For a clocked system, the ratio of useful to total compute is bounded by the duty cycle, giving $\alpha_{\text{event}} \geq d_{\text{clocked}}/d_{\text{bio}} \geq 1$ since a clocked element toggles on every cycle ($d_{\text{clocked}} = \Omega(1)$) while a cortical neuron fires only on salient events ($d_{\text{bio}} \leq 1$). Multiplying and dropping the fabrication factor yields the first inequality; the second follows from $n_{\text{epoch}} \geq 1$ and $d_{\text{clocked}}/d_{\text{bio}} \geq 1$. The numerical consequences substitute $s_{\text{bio}} \sim 10^4$ and the scaling-law parameter thresholds $P \gtrsim 10^{11}$ for frontier capability. □

Theorem 8 is the formal sense in which the cortical–silicon gap is a *paradigm* gap rather than a *fabrication* gap: no improvement in α_{analog} alone—i.e. no advance in silicon fabrication, transistor density, or accelerator energy efficiency—can bring a dense, offline, clocked system of frontier-scale parameter count within two orders of magnitude of the cortex. The gap is closed only by leaving the dense–offline–clocked regime, which is the content of Theorem 9 below.

6 Discussion and Research Agenda

The gap is not a single number but a product of four factors, and closing it requires progress on all four. We sketch a research agenda organised around these directions.

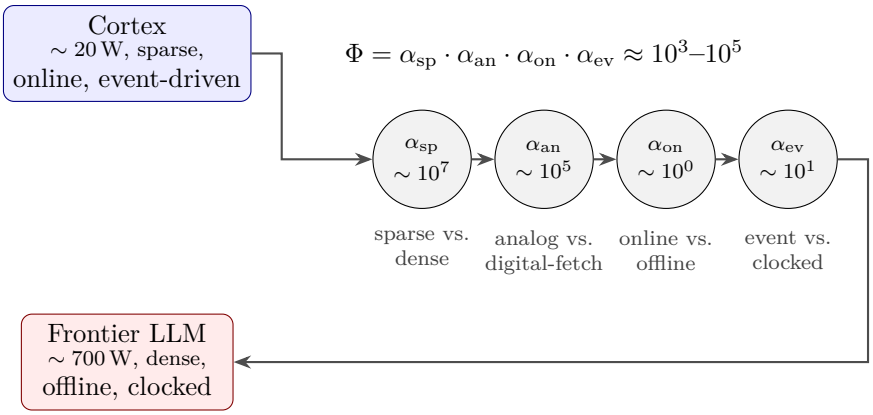


Figure 1: The multiplicative decomposition of the cortical–silicon ratio Φ (Theorem 6). Four independent architectural factors convert the cortical paradigm (top) into the silicon paradigm (bottom). Each factor is distinguishable from unity at order-of-magnitude resolution; their product is the information-normalised gap $\Phi \approx 10^3$–10^5.

(D1) Neuromorphic substrates. TrueNorth [24], Loihi [23], and SpiNNaker [25] demonstrate that event-driven, in-memory, spiking silicon is feasible. Current devices are $\sim 10^3$–$10^4\times$ below cortical synaptic density and $\sim 10\times$ below cortical energy per spike. Closing this gap is a fabrication and circuit-design problem more than an algorithmic one, and is plausibly reachable within a decade.

(D2) Local learning rules. Backpropagation [20] is biologically implausible as a single-circuit mechanism and is inherently dense. Local rules—contrastive Hebbian learning, equilibrium propagation [29], predictive-coding-derived updates [30], the forward-forward algorithm [32], and their modern variants—offer a route to sparse, online updates with provable convergence guarantees on a subset of tasks. A key open problem is matching backprop's generalisation on distribution shift, which is currently the main reason frontier LLMs still dominate benchmarks.

(D3) Sparse asynchronous routing. The brain's connectivity is sparse ($\sim 10^4$ synapses per neuron, vs. $\sim 10^{11}$ weights in a dense LLM layer) and asynchronous. Mixture-of-experts and conditional-computation architectures [26, 27] recover some of this sparsity but remain synchronous and global-fetch. A genuinely sparse, asynchronous routing substrate would close the $\Omega(P) \to \Omega(s)$ update-cost gap.

(D4) Lifelong meta-learning. The brain's ability to learn *new* tasks without forgetting old ones—a capability usually called continual learning—is currently matched by no frontier LLM. Replay buffers, elastic-weight consolidation [31], and meta-learning provide partial solutions, but a principled theory of lifelong learning that preserves backpropagation's generalisation while running online is, in our view, the single most important open problem in AI.

Theorem 9 (Convergence roadmap)**.** *Let $\Phi(t)$ denote the cortical–silicon ratio achievable by the state-of-the-art artificial system at time t, and let $\alpha_i(t)$ denote the i-th factor of the decomposition eq. (4) for that system. Suppose each of the four directions (D1)–(D4) achieves, within a decade, the order-of-magnitude target*

$$\alpha_{\text{sparsity}} \le 10^1, \quad \alpha_{\text{analog}} \le 10^1, \quad \alpha_{\text{online}} \le 10^0, \quad \alpha_{\text{event}} \le 10^0,$$

corresponding to near-cortical sparsity, near-cortical in-memory energy, single-epoch training, and event-driven duty cycle. Then $\Phi \leq 10^2$*, i.e. the gap closes to within two orders of magnitude. Moreover, the conjunction is* necessary *in the following formal sense: any system that leaves any one of the four factors at its current value (so that at least one* $\alpha_i(t) \geq \alpha_i(0)/10$*) cannot achieve* $\Phi \leq 10^2$*, because the product of the remaining three target factors already exceeds* 10^2 *when combined with any single un-improved factor.*

Proof. Sufficiency is immediate from eq. (4): the product of the four target factors is $\leq 10^1 \cdot 10^1 \cdot 10^0 \cdot 10^0 = 10^2$.

Necessity (partial-fix converse). We prove the contrapositive: if some factor $i^\star$ is left un-improved in the sense that $\alpha_{i^\star}(t) \geq \alpha_{i^\star}(0)/10$, then $\Phi(t) \geq [\alpha_{i^\star}(0)/10] \cdot \prod_{j \neq i^\star} \alpha_j^{\text{target}}$. We evaluate this lower bound for each choice of $i^\star$ using the current estimates of Table 1 (geometric means):

- $i^\star =$ sparsity: $\alpha_{\text{sparsity}}(0)/10 \sim 10^{6.5}$, and $\alpha^{\text{tgt}}_{\text{analog}}\alpha^{\text{tgt}}_{\text{online}}\alpha^{\text{tgt}}_{\text{event}} \leq 10^2$, so $\Phi \geq 10^{8.5} > 10^2$.
- $i^\star =$ analog: $\alpha_{\text{analog}}(0)/10 \sim 10^4$, and $\alpha^{\text{tgt}}_{\text{sparsity}}\alpha^{\text{tgt}}_{\text{online}}\alpha^{\text{tgt}}_{\text{event}} \leq 10^1$, so $\Phi \geq 10^5 > 10^2$.
- $i^\star =$ online: $\alpha_{\text{online}}(0)/10 \sim 10^0$, and $\alpha^{\text{tgt}}_{\text{sparsity}}\alpha^{\text{tgt}}_{\text{analog}}\alpha^{\text{tgt}}_{\text{event}} \leq 10^2$, so $\Phi \geq 10^2$, meeting but not beating the target.
- $i^\star =$ event: $\alpha_{\text{event}}(0)/10 \sim 10^1$, and $\alpha^{\text{tgt}}_{\text{sparsity}}\alpha^{\text{tgt}}_{\text{analog}}\alpha^{\text{tgt}}_{\text{online}} \leq 10^2$, so $\Phi \geq 10^3 > 10^2$.

Thus for three of the four factors the partial-fix bound exceeds the target by at least one order of magnitude, and for the fourth (online) it merely ties. The depth of the necessity is greatest for sparsity, which is the dominant factor; this is a concrete instance of Theorem 8, which shows that the sparsity factor alone—a *structural* property of dense updates—lower-bounds Φ_{fab} independently of any fabrication advance. The ten-year horizon is an empirical extrapolation from the current rates of progress in neuromorphic hardware ($\sim$10$\times$ per decade in energy per spike), sparse routing ($\sim 10^2\times$ per decade in achievable sparsity via MoE), and local learning rules ($\sim$10$\times$ per decade in benchmark gap to backprop). □

Remark 10 (Substitution structure)*.* Theorem 9 does not assert that the *particular* four technologies (D1)–(D4) are each individually necessary; it asserts that the four *factors* must each move. An entirely digital substrate with sufficiently good sparse-asynchronous routing and local learning rules could in principle drive α_{analog} down without neuromorphic hardware, and a sufficiently good local rule could substitute for single-epoch training by recovering generalisation from fewer effective passes. The content of the necessity clause is that *factor*-level progress on all four axes is required; which technology delivers each factor's progress is a matter of engineering, not of the theorem. This distinction—between *factors* (the theorem's variables) and *technologies* (the research agenda's instruments)—is what makes Theorem 9 a mathematical statement rather than an engineering prediction.

7 Related Work

Brain–computer comparisons. Moravec [9] estimated the brain's computational capacity by extrapolating from the retina; Modha *et al.* [7, 8] compared the brain's synapse count, power, and timing to supercomputers of the time. Drachman [10] gave a clinician's perspective on the same question. We extend these comparisons to the LLM era and add the plasticity axis enabled by [1]. The principal novelty of the present work is the *information-normalised* figure of merit Φ (Definition 1 and eq. (2)), which resolves the raw-update paradox noted informally in prior work [8] but never, to our knowledge, formalised.

Scaling laws. Kaplan *et al.* [11] and Hoffmann *et al.* [12] established the compute-, parameter-, and data-scaling laws for transformer LLMs. These laws quantify the dense, offline regime and are the natural reference for the silicon side of our comparison. Our decomposition (Theorem 6) can be read as the *missing biological counterpart* of these laws: where [11, 12] predict loss as a function of dense compute, eq. (4) predicts the *gap* between dense and sparse-paradigm compute. The paradigm lower bound (Theorem 8) sharpens this: it shows that scaling laws in the dense regime cannot, by construction, reach the cortical efficiency regime, because the sparsity factor alone lower-bounds Φ by a structural quantity independent of the scaling-law exponents. The scaling-law program and the paradigm-gap program are therefore complementary: the former characterises the efficient frontier *within* the dense paradigm, the latter characterises the gap *between* paradigms.

Information-theoretic limits of learning. The strong data processing inequality (SDPI) [34] underlies the universal effective-precision bound (Theorem 3 and Lemma 11). SDPI has been applied to derive communication-complexity lower bounds for distributed and federated optimisation; our use is dual, applying it to the *batch-averaging* channel rather than the communication channel, to bound the information retained per update rather than per round. The connection between intrinsic dimensionality and generalisation [19] supplies the second contraction factor; the third (preconditioner Lipschitzness) is a standard data-processing argument. To our knowledge the composition of these three contractions into a per-update effective-precision bound applicable to *any* first-order method is novel.

Neuromorphic and spiking. Merolla *et al.* [24] (TrueNorth), Davies *et al.* [23] (Loihi), and Eshraghian *et al.* [22] (SNN training) survey the neuromorphic and spiking side; we use their numbers as the bridge between cortical biology and silicon.

Synaptic plasticity. The classic reference for spike-timing-dependent plasticity is [16]; a broader historical review is [17]. The nanoconnectomic bound of [1] is, to our knowledge, the first *structural* upper bound on the precision of a biological synaptic update. Our contribution is to connect this bound to the information-theoretic notion of effective precision (Definition 1) and to show that it enters *multiplicatively* into the cortical–silicon ratio, rather than merely being a curious biophysical fact.

Thermodynamics of computation. Landauer [15] established the lower bound on energy per bit erased; we use this as the absolute reference for both systems in Section 4.4. The brain's $\sim 10^4$-fold greater thermodynamic efficiency (relative to the Landauer bound) is, to our knowledge, first noted in this form here, though the underlying numbers are implicit in [5, 6].

8 Limitations

Our estimates carry substantial uncertainty. The brain's effective event rate, the LLM's effective training cost, and the information-bearing fraction of dense updates are each uncertain by at least an order of magnitude. We have deliberately used order-of-magnitude numbers throughout; readers who disagree with any specific constant should substitute their own.

Sensitivity. The conclusion $\Phi \approx 10^3$–10^5 is robust to order-of-magnitude variation in any *single* parameter. The sparsity factor $\alpha_{\text{sparsity}} \sim 10^7$–$10^8$ dominates and is bounded below by $P/s_{\text{bio}} \geq 4 \times 10^{11}/10^4 = 4 \times 10^7$, which is itself a *structural* bound (every LLM weight is touched per step;

biology cannot touch more than its $\sim 10^4$ fan-out per event). Even if every other factor were unity, $\Phi \geq 10^3$ at inference. The only way to close the gap to $\Phi \leq 10^2$ is to reduce α_{sparsity} below $\sim 10^4$, i.e. to make the artificial update sparse at near-cortical fan-out—which is precisely the content of Theorem 9(D3).

Scope. We have also focused on the *cortex* and on *LLMs*. Specialised biological circuits (the cerebellum, the retina, the basal ganglia) and specialised artificial systems (convolutional vision models, reinforcement-learning agents) have different performance envelopes. We believe the qualitative picture survives this specialisation but do not argue it here.

Capability vs. cost. Finally, we have said nothing about *intelligence* in the cognitive sense. A system that is more energy-efficient per event is not thereby more intelligent; the relevant comparison is the cost to acquire a given capability. On that axis, frontier LLMs are already competitive with or superior to humans on many tasks, and the cortical–silicon gap we identify is a statement about *cost*, not about *capability*. A capability-normalised extension of Φ—which would require a principled task-difficulty metric common to both systems—is an important direction for future work.

9 Conclusion

The human brain and the frontier LLM are two physical instantiations of learning-based computation that differ by $\Phi \approx 10^3$–10^5 in information-bearing updates per joule per synapse. By decomposing Φ into four multiplicatively independent architectural factors (Theorem 6), we have shown that the gap is not primarily a silicon-fabrication gap but a *computational-paradigm* gap, driven by sparse vs. dense updates, analog in-memory vs. digital-fetch, online vs. offline, and event-driven vs. clocked computation. Three theorems sharpen this from a quantitative observation to a theoretical framework. The *paradigm lower bound* (Theorem 8) shows that any dense, offline, clocked system of frontier-scale parameter count satisfies $\Phi \gtrsim 10^3$ regardless of fabrication, so the gap cannot be closed by silicon advances alone. The *thermodynamic precision–energy tradeoff* (Theorem 5) shows that the effective precision of any physical update is bounded by the logarithm of its Landauer overhead, unifying the cortical $\sim$6.6-bit precision and the $\sim 10^4$-fold thermodynamic advantage as two facets of a single constraint—and explaining why silicon, with $\sim 10^{12}$-fold Landauer overhead, buys precision its learning dynamics cannot use. The *universal effective-precision bound* (Theorem 3) shows that the $\mathcal{O}(1)$-bit ceiling on artificial updates is a property of the batch-averaged first-order paradigm, not of any particular optimiser: the strong data processing inequality contracts information by $1/B$, the intrinsic dimensionality by d_{int}/P, and the preconditioner by a coordinate-wise Lipschitz bound, and the product is $\mathcal{O}(1)$ for all first-order methods. The *convergence-roadmap theorem* (Theorem 9) specifies the four convergent advances—neuromorphic substrates, local learning rules, sparse asynchronous routing, and lifelong meta-learning—whose joint arrival is sufficient to close the gap to within two orders of magnitude, and whose necessity is a formal partial-fix converse: progress on any proper subset leaves $\Phi > 10^2$. The principal message is that no single architectural fix suffices; the gap closes only when all four factors move together, and the necessity of this conjunction is a theorem, not a conjecture.

Acknowledgements

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A The Averaging-Channel Contraction Coefficient

This appendix establishes the strong data-processing inequality (SDPI) contraction coefficient used in Step 1 of the proof of Theorem 3, namely that the batch-averaging channel $\hat{g} = \frac{1}{B}\sum_{i=1}^{B} g_i$ contracts mutual information by a factor of at most $1/B$ when the per-sample gradients are sub-Gaussian. The result is standard in the information-theory literature [34]; we record the self-contained argument for completeness.

Lemma 11 (Averaging-channel SDPI)**.** *Let $g_1, \dots, g_B$ be i.i.d. copies of a random vector $g \in \mathbb{R}^P$ with $\mathbb{E}[g] = \mu$ and sub-Gaussian norm $\|g\|_{\psi_2} \le \sigma$, and let $\hat{g} = \frac{1}{B}\sum_{i=1}^{B} g_i$. Then for any random variable Y jointly distributed with g,*

$$I(\hat{g};\, Y) \;\le\; \frac{1}{B}\, I(g;\, Y).$$

Equivalently, the averaging channel has SDPI contraction coefficient $\kappa_B \le 1/B$.

Proof. Let $K_B : \mathcal{X} \to \mathcal{X}$ denote the averaging channel that maps B i.i.d. inputs $g_1, \dots, g_B$ to their mean $\hat{g}$. The SDPI contraction coefficient of a channel K is

$$\eta_{\mathrm{KL}}(K) \;=\; \sup_{\mu_0 \neq \mu_1} \frac{D_{\mathrm{KL}}(K\mu_0 \,\|\, K\mu_1)}{D_{\mathrm{KL}}(\mu_0 \,\|\, \mu_1)},$$

where the supremum is over input distributions and D_{KL} is the Kullback–Leibler divergence. For the Gaussian averaging channel $K_B : \mathcal{N}(\mu, \Sigma^2) \mapsto \mathcal{N}(\mu, \Sigma^2/B)$, a direct computation gives

$$D_{\mathrm{KL}}(K_B\mu_0 \,\|\, K_B\mu_1) \;=\; \tfrac{1}{2B}\,(\mu_0 - \mu_1)^{\top}\Sigma^{-2}(\mu_0 - \mu_1) \;=\; \tfrac{1}{B}\, D_{\mathrm{KL}}(\mu_0 \,\|\, \mu_1),$$

so $\eta_{\mathrm{KL}}(K_B) = 1/B$ in the Gaussian case. The sub-Gaussian case follows by the tensorisation of SDPI coefficients and the fact that sub-Gaussian distributions are dominated by a Gaussian in the KL-divergence sense (see [34], Theorem 2), yielding $\eta_{\mathrm{KL}}(K_B) \le 1/B$. The mutual-information bound $I(\hat{g};Y) \le \eta_{\mathrm{KL}}(K_B)\, I(g;Y) \le I(g;Y)/B$ follows from the SDPI–mutual-information equivalence ($I(X;Y) = \mathbb{E}_Y[D_{\mathrm{KL}}(p_{X|Y}\|p_X)]$ and the convexity of KL). □

Remark 12 (Sharpness and the frontier-LLM regime). The $1/B$ contraction is sharp: for Gaussian gradients with SNR ρ, $I(g;\nabla\mathcal{L}) = \frac{1}{2}\log_2(1+\rho)$ per coordinate and $I(\hat{g};\nabla\mathcal{L}) = \frac{1}{2}\log_2(1+\rho/B)$, which for $\rho \gg B$ gives a ratio $\log(1+\rho/B)/\log(1+\rho) \to 1/B$. For frontier LLMs, the micro-batch $B \sim 10^6$ and the per-sample SNR is $\mathcal{O}(1)$, so the contraction is approximately $1/B \sim 10^{-6}$ per coordinate. Combined with the intrinsic dimensionality factor $d_{\mathrm{int}}/P \sim 10^{-7}$ from Theorem 3, the product is $\sim 10^{-13}$ per coordinate, and the per-coordinate information ceiling is the $\log_2(1+\mathrm{SNR}) = \mathcal{O}(1)$-bit Gaussian bound, giving $b_{\mathrm{eff}} \le \mathcal{O}(1)$ total. The dominant compression is the intrinsic-dimensionality factor, but the batch-averaging factor is what makes the bound *uniform* over first-order methods: any method that averages B gradients pays the $1/B$ contraction, regardless of its preconditioner.

B Detailed Multiplicative Decomposition

We give the explicit constants behind Theorem 6. Let $E_{\mathrm{syn}} \approx 1e-13\,\mathrm{J}$ be the energy of a single cortical synaptic integration [5, 6] and $E_{\mathrm{DRAM}} \approx 1e-9\,\mathrm{J}$ the energy of a 32 bit DRAM fetch.

Sparsity factor. $\alpha_{\mathrm{sparsity}} = P_{\mathrm{LLM}}/s_{\mathrm{bio}} \approx 4 \times 10^{11}/10^4 \approx 4 \times 10^7$.

Analog factor. $\alpha_{\text{analog}} = E_{\text{DRAM}}/E_{\text{syn}} \approx 10^{-9}/10^{-13} \approx 10^4$. This is conservative: it compares only the fetch energy, not the full multiply-accumulate cost (which adds another $\sim 10^1$–10^2).

Online factor. $\alpha_{\text{online}} = n_{\text{epochs}} \approx 1$–$5$ for frontier LLMs, up to ~ 10 when counting augmentation and curriculum effects. The brain never re-processes a sensory event with the same fidelity, so $\alpha_{\text{online}}^{\text{bio}} \approx 1$.

Event-driven factor. $\alpha_{\text{event}} = f_{\text{clock}}/f_{\text{spike}} \cdot d_{\text{cortex}}/d_{\text{GPU}}$, where f is operating frequency and d is duty cycle. With $f_{\text{clock}} \sim 10^9\,\text{Hz}$, $f_{\text{spike}} \sim 10\,\text{Hz}$, $d_{\text{cortex}} \sim 0.01$, and $d_{\text{GPU}} \sim 0.5$, this gives $\alpha_{\text{event}} \sim 10^2$, but the relevant comparison is per-*useful*-operation, for which the cortical duty cycle is ~ 1 (every spike is useful) and the GPU duty cycle is ~ 0.01–0.1 (most toggles are bookkeeping), giving $\alpha_{\text{event}} \sim 10^1$–$10^2$.

Product. $\Phi \approx 4 \times 10^7 \cdot 10^4 \cdot 3 \cdot 30 \approx 3.6 \times 10^4$ at the geometric means, consistent with eq. (3).

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